

RESEARCH PROFILE

I am a roboticist specializing in **real-to-sim**. I turn real-world observations — a robot arm pushing a box, a soft body deforming, a magnetic guidewire bending — into physically accurate digital twins that a simulator can replay, and use them to train and evaluate policies for real robots. My work spans agentic world modeling with vision-language models, soft-body system identification from video, and Cosserat-rod modeling of slender magnetic structures, together with the simulators and hardware built to close the **sim-to-real** loop. I also study world models and vision-language-action (VLA) models for robot learning.

EDUCATION

University of British Columbia Sep 2026 – Present
Ph.D. in Computer Science
Advisor: Peter Yichen Chen — UBC PhysAI Lab

Columbia University Sep 2024 – May 2026
M.S. in Mechanical Engineering, Research Track — Concentration: Robotics and Control
Advisor: Hod Lipson — Creative Machines Lab

Shanghai Jiao Tong University Sep 2020 – Jun 2024
B.S. in Mechanical Engineering; double major in Mathematics and Applied Mathematics

PUBLICATIONS

- [1] G. Chen*, Y. Qu*, **Q. Xia***, P. Jing, Y. Cheng, B. Ma, P. Yang, B. Zhou, Z. Li, S. Suri, G. Sun, C. Liu, P. Y. Chen, Z. Zeng, and F. Shi. DiagGen: Agentic Generation of Deformable Assets with Sim-based Diagnostics for Robotic Simulation. Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. (*equal contribution) [\[arXiv:2609.23103\]](#) [\[project\]](#)
- [2] G. Chen*, **Q. Xia***, J. Peng*, H. Zhang*, P. Jing*, B. Ma*, J. Qian, Y. Cheng, Z. Jiao, B. Zhou, Y. Qu, L. Ye, K. Zhang, K. Wang, W. Zeng, Y. Chen, P. Yang, Z. Zeng, S. Luo, H. Wang, C. Liu, A. Yuille, F. Shi, C. Zheng, Y. Li, C. Jiang, and P. Y. Chen. Agentic Real2Sim: Physics-based World Modeling with Vision-Language Agents. Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. (*equal contribution) [\[arXiv:2607.19190\]](#) [\[project\]](#) [\[code\]](#)
- [3] J. Lin*, **Q. Xia***, S. Zhang, and H. Lipson. RoboVoxel: Learning Soft Body Dynamics via Video Colorization. *Manuscript in preparation for journal submission*, 2026. (*equal contribution)
- [4] H. Liu, **Q. Xia**, R. Wang, C. Huang, Z. Ullah, K. Xu, and D. Wang. An MLP-Embedded Transformer-Physics Hybrid Framework for Inverse Design of Magnetic Rod. Under review at *IEEE Transactions on Automation Science and Engineering (T-ASE)*, 2026.

RESEARCH EXPERIENCE

Agentic Real2Sim: Physics-based World Modeling with Vision-Language Agents Mar 2026 – Sep 2026
[\[project page\]](#)

University of British Columbia — PhysAI Lab. Advisor: Peter Yichen Chen

- Co-developed an agentic framework in which **vision-language agents** convert recordings of real robot-object interaction into **simulatable digital twins**, recovering scene geometry, object state, and physical parameters without per-scene hand tuning.
- One unified Real2Sim pipeline covers rigid-object manipulation, **deformable-object interaction**, and **humanoid motion** — cases that previously required separate, specialized Real2Sim methods.
- Resulting digital twins support downstream **policy learning** and policy evaluation in simulation; an **open-weight VLM** backend matches frontier-model success rates, making the pipeline cheap enough to run at dataset scale.

DiagGen: Agentic Generation of Deformable Assets with Sim-based Diagnostics Jun 2026 – Sep 2026
[\[project page\]](#)

University of British Columbia — PhysAI Lab. Advisor: Peter Yichen Chen

- Improved the physical fidelity and stability of generated deformable assets in the **physics simulator** through systematic calibration of material and solver parameters and the introduction of safeguard bounds, yielding more plausible and robust object behavior under robotic manipulation.
- Integrated DiagGen into the Agentic **Real2Sim** pipeline, enabling end-to-end generation of **simulation-ready scenes** directly from videos of a robot arm manipulating deformable objects.

RoboVoxel: Soft-Body System Identification from Videos [\[project page\]](#) Mar 2025 – Present
Columbia University — Creative Machines Lab. Advisors: Jiong Lin, Hod Lipson

- Built a physics-based **soft-body simulation** environment (elastic bodies under gravity and actuator forces) and used it to generate large-scale synthetic datasets spanning diverse geometries, material parameters, and actuation patterns to supervise video-based **system identification**.
- Trained a **Video Transformer** that infers voxel-level soft-body physical parameters — stiffness, mass, and actuator layout — from multi-frame video, enabling video-driven digital-twin reconstruction that a simulator can directly replay.
- Built a 4×4 physical voxel robot — 16 independently servo-driven active voxels with a gear-and-curved-rack mechanism — mirroring the simulator's actuation, and validated the model's **sim-to-real** system identification on real videos of the robot.

An MLP-Embedded Transformer-Physics Hybrid Framework for Inverse Design of Magnetic Rod [\[project page\]](#) Sep 2023 – Jun 2025

Shanghai Jiao Tong University. Advisor: Dong Wang. Manuscript in submission.

- Built a **Cosserat-rod** simulator that captures the full nonlinear dynamics of slender elastic structures — bending, twisting and shear — a general-purpose framework directly transferable to **cable, rope, and continuum-robot manipulation**.
- Extended the simulator with external magnetic body-torques to model a **magnetically actuated vascular guidewire**; Cosserat-rod predictions matched physical prototype behavior under a range of actuation profiles, enabling **model-based control** and sim-based policy design.
- Prototyped the hardware loop for **closed-loop validation**: a robot arm positions the driving magnet while a motor rotates the guidewire base; full control stack implemented in Python with SolidWorks-designed hardware and Xbox-controller teleoperation.

PROJECTS

Rise of the AI Knight — World-Model RL for Action-Platformer Games [\[project page\]](#) Sep 2025 – Present

Columbia University. Advisor: Shipra Agrawal

- Implemented a **world-model-based reinforcement-learning** agent for Hollow Knight and Hollow Knight: Silksong, adapting **latent-dynamics (Dreamer-style)** methods to a complex, real-time action-platformer environment with long-horizon credit assignment.

M.E.H: A Bipedal Robot [\[project page\]](#)

Mar 2025 – May 2025

Columbia University. Advisor: Hod Lipson

- Designed a parallel-linked legged robot and derived closed-form **inverse kinematics** for **gait planning**.
- Deployed a **real-time locomotion controller** on a Raspberry Pi, reaching a maximum walking speed of approximately 32 cm/s.

INDUSTRY EXPERIENCE

Shanghai ABB Engineering Co., Ltd

Jun 2023 – Sep 2023

Computer Vision Intern

- Developed and tested **computer-vision** algorithms for industrial robotic automation alongside the perception team.
- Built a binocular-stereo vision demo in Halcon for automated tire-assembly positioning, achieving **1 mm 3D localization accuracy** — a real-to-sim-style calibration task bridging CAD models and camera observations.

SKILLS

- **Simulation & Physics:** MuJoCo, Taichi, differentiable physics, soft-body dynamics, cable / deformable-object simulation
- **Robotics & ML:** Video Transformers, VLA models, model-based reinforcement learning (world models), robot kinematics & dynamics
- **CAD & CAE:** SolidWorks, Fusion360, UG, Catia, Adams, Ansys